

Seminaropgave

Anvendt Corporate Finance

Currency Risk Hedging at Sandvik: Does the Choice of Risk Measure Change the Optimal Hedging Strategy?

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Abstract

This paper examines whether the choice between Value-at-Risk (VaR) and Expected Shortfall (ES) as the primary risk measure leads to materially different hedging recommendations for Sandvik Group's foreign-denominated net assets. We apply Monte Carlo simulations under both Gaussian and Student- t copula specifications to model translation risk across Sandvik's major currency exposures over a one-year horizon. Hedging strategies are derived from two optimisation frameworks: a VaR-minimising framework and a CVaR-minimising framework. The results show that the choice of risk measure can materially affect hedge ratios when currency returns exhibit fat tails and tail dependence. Under the Student- t copula, the ES/VaR ratio is 1.286, exceeding the Gaussian benchmark of approximately 1.254, which indicates that the portfolio contains tail risk beyond what a normal model would imply. The largest divergence between the VaR- and CVaR-optimal strategies occurs at moderate levels of cost sensitivity. At $\lambda = 7$, the VaR-optimal strategy retains only full EUR and full CNY hedging, having dropped USD entirely, while the CVaR-optimal strategy additionally retains 88% USD coverage. As a result, the CVaR-optimal strategy reduces residual ES from SEK 5.929 billion to SEK 2.076 billion, although at a higher annual hedge cost. These findings suggest that, for multinational firms with fat-tailed and correlated currency exposures, the choice of risk measure is not merely a technical modelling decision, but can materially influence corporate hedging policy.

Contributions

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1 Introduction

The Fundamental Review of the Trading Book (FRTB) represents a central part of the post-crisis reform of bank market-risk regulation. One of its most important changes is the replacement of Value-at-Risk (VaR) with Expected Shortfall (ES) under the internal models approach for bank market-risk capital (Basel Committee on Banking Supervision, 2019). Although the European Union implemented most Basel III requirements from January 2025, the application of the FRTB market-risk capital requirements has been postponed to January 2027 (European Commission, 2025). The reform nevertheless reflects a broader regulatory shift toward tail-sensitive risk measurement. This shift is motivated by two central limitations of VaR: it does not always capture diversification properly, and it provides no information about the severity of losses once the confidence threshold has been breached. These limitations were already established before the financial crisis, Yamai and Yoshida (2005) show that VaR can be misleading in fat-tailed settings because it ignores the magnitude of losses beyond the selected quantile. The 2008 financial crisis helped highlight this critique and contributed to the post-crisis shift toward risk measures that capture tail severity rather than only loss thresholds. Although the FRTB is designed for banks rather than non-financial firms, the underlying critique is relevant for corporate risk management. If exchange-rate returns are fat-tailed and dependent in stress periods, VaR-based hedging strategies may leave firms underprotected precisely when downside protection is most valuable.

This paper examines whether the choice between VaR and ES leads to materially different hedging recommendations for Sandvik Group's foreign-denominated net assets. The research question is:

Does the choice between Value-at-Risk and Expected Shortfall as the primary risk measure lead to materially different hedging recommendations for Sandvik's foreign-denominated net assets, and if so, what are the implications for corporate treasury risk management?

Sandvik is a Swedish multinational engineering group that reports in SEK while holding SEK 101.18 billion in foreign-denominated net assets. The five largest individually disclosed currency positions, in EUR, USD, AUD, GBP, and CNY, amount to SEK 82.74 billion, or 81.8% of the total. Translation exposure of this size can generate Other Comprehensive Income (OCI) adjustments large enough to affect reported equity ratios, covenant headroom, and external credit

assessments, even if the underlying losses are unrealised in a cash-flow sense. Sandvik therefore provides a useful empirical setting for analysing whether the choice of risk measure can alter corporate hedging policy.

To answer the research question, we construct a Monte Carlo simulation model of Sandvik's five largest currency exposures over a one-year horizon. We compare two distributional specifications: a Gaussian copula benchmark and a Student- t copula calibrated to empirical return data, allowing for fat-tailed marginals and tail dependence. For each specification, we estimate VaR and ES at the 95% confidence level for the unhedged portfolio and for alternative hedging strategies. Hedge ratios are then derived from two optimisation frameworks: a VaR-minimising framework and a CVaR-minimising framework based on Rockafellar and Uryasev (2000). Since CVaR is equivalent to ES under the continuous loss distributions used in the simulation, the comparison allows us to assess whether targeting the loss threshold or the average loss beyond that threshold changes the recommended hedge ratios and the residual downside risk.

The paper contributes in three ways. First, it applies the ES framework to corporate translation risk, an area where VaR and variance-based approaches remain more common. Second, it provides a direct comparison of VaR-optimal and CVaR-optimal hedge ratios for the same firm, exposure set, and cost structure. Third, it examines whether fat tails and tail dependence materially affect hedging recommendations by comparing Gaussian and Student- t Monte Carlo results. The analysis therefore links a regulatory debate in market-risk measurement to a practical corporate treasury problem.

2 Literature Review

This section reviews the literature most relevant to the analysis. It first outlines the main forms of currency risk faced by multinational corporations, with particular emphasis on translation exposure. It then reviews the shift from Value-at-Risk to Expected Shortfall in both the academic and regulatory literature, and finally discusses how the choice of risk measure affects hedging and portfolio optimization.

2.1 Currency Risk in Multinational Corporations

Multinational corporations face currency risk because they operate, invest, and hold assets across countries with different currencies. Exchange rate movements can therefore affect reported earnings, balance sheet values, and competitive position. The literature typically distinguishes between three forms of currency exposure: transaction exposure, translation exposure, and economic exposure (Madura, [2020](#)).

Transaction exposure refers to the effect of exchange rate movements on contracted cash flows, such as receivables and payables denominated in foreign currencies. This exposure is usually short- to medium-term and directly affects realised profits. Economic exposure is broader and more long-term, capturing how exchange rate changes affect a firm's competitive position, future revenues, and overall market value (Adler & Dumas, [1984](#)). Translation exposure arises when the financial statements of foreign subsidiaries are converted into the parent company's reporting currency. Under current accounting standards, these adjustments flow through Other Comprehensive Income (OCI) rather than the income statement (Papaioannou, [2006](#)).

A Sandvik-specific applied study is provided by Carlsson and Ericsson ([2012](#)), who examine the company's Layered Basket Option hedging strategy. Their analysis focuses on option-based hedging of currency exposure rather than translation exposure specifically, but it is relevant because it documents how Sandvik has historically approached FX risk management in practice. The study emphasises that Sandvik's hedging objective was not to speculate on currency movements, but to reduce FX-driven volatility and produce more predictable results for stakeholders. It also highlights the practical importance of currency weights and correlations, since basket hedging only creates diversification benefits when the structure of the currency basket is economically sensible.

This paper focuses on translation exposure. Although it is often treated as an accounting issue with no direct cash flow effect, that view understates its relevance. OCI adjustments affect reported equity, and for firms with covenants linked to equity ratios or book value, large translation losses can have real consequences even if they are unrealised. Balance sheet volatility may also matter for external credit assessments and investor perceptions. For Sandvik, which holds substantial net assets in EUR, USD, AUD, GBP, and CNY, the potential size of these adjustments justifies formal risk measurement and hedging analysis.

2.2 From VaR to Expected Shortfall: Theory, Regulation, and Optimization

Value-at-Risk became the dominant market risk measure after J.P. Morgan's RiskMetrics framework (J.P. Morgan, 1996), and it remains widely used in both financial and corporate practice (Jorion, 2007). However, the literature identifies two important limitations. First, VaR does not always capture diversification properly (Artzner et al., 1999). Second, it provides no information about the size of losses once the confidence threshold has been exceeded. Yamai and Yoshida (2005) show that this can make VaR misleading in settings with fat tails or stress events, where tail losses matter most.

Expected Shortfall addresses these weaknesses by measuring the average loss conditional on losses exceeding the VaR threshold. Acerbi and Tasche (2002) show that ES has stronger theoretical properties than VaR, and these arguments helped motivate the Basel Committee's Fundamental Review of the Trading Book (FRTB). Although this regulatory change does not apply to non-financial corporates, the underlying logic is still relevant for any portfolio of correlated currency exposures.

The choice of risk measure also affects hedging design. The standard benchmark is the mean-variance framework of Markowitz (1952), which can be extended to include hedging costs (Kolm et al., 2014). Its limitation is that variance treats upside and downside deviations symmetrically. Applied evidence from Sandvik also supports the relevance of portfolio-based hedging logic: Carlsson and Ericsson (2012) show that the effectiveness of basket hedging depends on the interaction between currency weights, volatilities, and correlations. This is consistent with the broader portfolio-theory argument that the risk contribution of a currency cannot be assessed only from its standalone exposure.

By contrast, Rockafellar and Uryasev (2000) show that CVaR, which is equivalent to ES, can be minimised directly, and Rockafellar and Uryasev (2002) show that the problem can be reformulated as a linear program. This makes CVaR-based hedging tractable even in large Monte Carlo settings.

Under normality, VaR and CVaR tend to produce similar hedge rankings because second moments describe the distribution sufficiently well. Under fat tails, however, the results may differ

materially. This paper therefore extends the Sandvik-specific hedging literature by shifting the focus from option-based basket hedging and premium reduction to the question of whether VaR- and ES-based optimisation produce different hedge ratios for Sandvik’s foreign-denominated net assets.

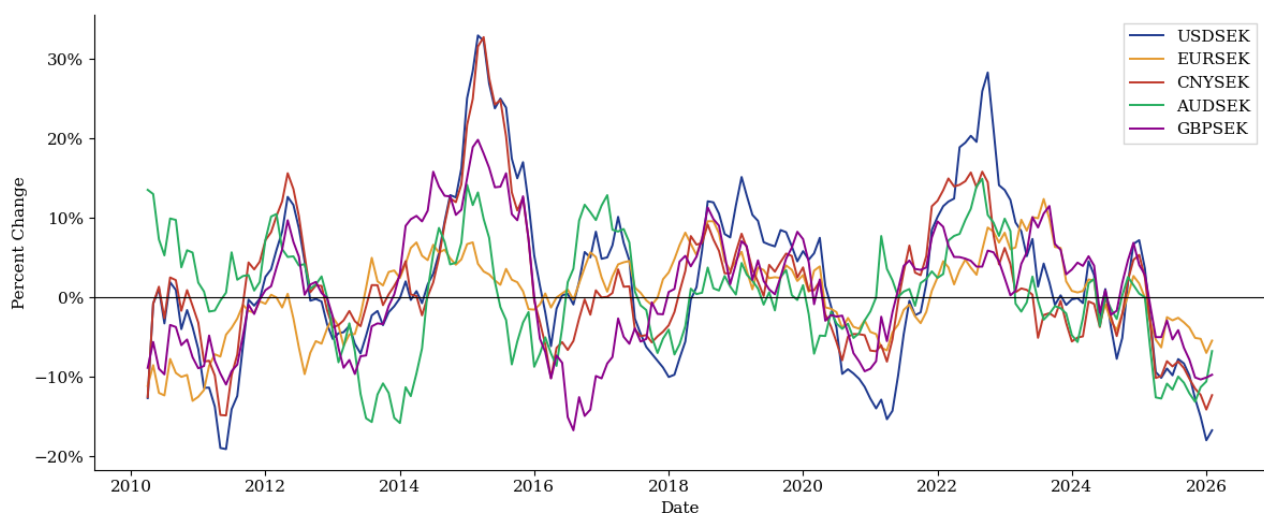
3 Hedging Currency Risk

This section examines the key sources of currency risk faced by Sandvik and outlines the financial instruments available to manage this exposure. In particular, it focuses on the volatility of relevant exchange rates and the use of forward contracts as a practical tool for hedging foreign-denominated net assets.

3.1 Volatility in Currency Markets

Currency markets have experienced considerable volatility in recent years. The COVID-19 pandemic, the war in Ukraine, and divergent monetary policy responses across major central banks have all contributed to large and sometimes abrupt exchange rate movements. Figure 1 shows the rolling 12-month percentage changes of Sandvik’s key currency pairs against SEK from 2010 to 2026.

Figure 1: Rolling 12-month changes in selected currencies relative to SEK.



Source: Riksbank and own calculations made on the 23rd of March 2026.

As the figure illustrates, annual currency swings of 10% or more against SEK are not unusual, and several of Sandvik’s exposure currencies have experienced moves well beyond that threshold within the sample period. For a firm holding tens of billions of SEK in foreign net assets, such

fluctuations can translate into material OCI adjustments. This volatility provides the practical motivation for the hedging analysis that follows: even if translation losses are unrealised in a cash flow sense, they affect reported equity and can be large relative to annual earnings.

3.2 Hedging Instruments: Forward Contracts

The most common instrument for hedging translation exposure is the currency forward contract. A forward is a binding agreement between two parties to exchange a specified amount of currency at a predetermined rate K on a future date T . At maturity, the value of a long forward position is

$$V_K(T, T) = S_T - K \quad (1)$$

where S_T is the spot rate at maturity (Hull, 2015). By locking in a conversion rate today, the firm eliminates uncertainty about the future SEK value of its foreign net assets, but it also gives up any benefit from favourable exchange rate movements.

The cost of a forward hedge can be approximated by the difference between the current spot rate and the forward rate, $S_0 - F_{1Y}$, which reflects the interest rate differential between the two currencies.¹ A positive hedging cost means the forward rate is less favourable than the current spot rate, so the firm pays a premium for certainty. The size of this premium varies across currencies and is an important input to the hedge ratio optimisation in Section 7.

Currency options offer an alternative: they provide the right, but not the obligation, to exchange at a preset rate, which means the firm can protect against downside movements while retaining upside potential. However, options require an upfront premium and are less commonly used for translation hedging, where the objective is typically to reduce balance sheet volatility rather than to preserve optionality (Papaioannou, 2006). This paper therefore focuses exclusively on forward contracts as the hedging instrument.

4 Sandvik Group

This section introduces Sandvik Group as the empirical setting for the analysis. It first provides a brief company overview, then describes the firm's foreign-denominated net assets and resulting

¹This is a simplification. The true cost of hedging can only be observed ex post and also depends on bid-ask spreads, credit charges, and liquidity conditions. We discuss this limitation in Section 8.4.

currency exposure, and finally presents preliminary evidence on the distributional properties of Sandvik's FX returns.

4.1 Company Overview

Sandvik Group is a Swedish multinational engineering company headquartered in Stockholm. Founded in 1862, the Group has developed into a global supplier of equipment, tools, services, and industrial solutions, with operations and customers across a wide range of international markets. Due to this broad geographic presence, Sandvik is significantly exposed to foreign currencies through its foreign sales, operations, and balance sheet items.

For this paper, Sandvik is a relevant case for analyzing currency risk because the company reports in SEK while a substantial part of its net assets is denominated in other currencies. This creates translation exposure, as exchange rate movements affect the SEK value of foreign net assets when they are consolidated in the financial statements. Sandvik's international structure therefore makes it well suited for examining how the choice of risk measure may influence the optimal hedging strategy.

A previous Sandvik-specific study provides useful context for the Group's historical approach to currency risk management. Based on personal communication with Thomas Hamberg, then Head of Corporate Finance at Sandvik, Carlsson and Ericsson (2012) report that Sandvik's hedging objective was not to generate speculative gains, but to reduce the volatility caused by currency fluctuations and produce more predictable outcomes for stakeholders. This suggests a relatively conservative hedging philosophy, where the treasury function is primarily concerned with limiting unexpected FX-driven deviations, even if hedging contracts may reduce the potential benefits from favourable currency movements. While this evidence concerns Sandvik's hedging practice in 2012 and should not be interpreted as a current formal risk appetite statement, it supports the view that FX risk management is operationally relevant for Sandvik and that hedge design should be evaluated against both risk reduction and implementation cost.

4.2 Foreign-Denominated Net Assets and Currency Exposure

As shown in Table 1, Sandvik's foreign-denominated net assets are concentrated in a few major currencies. The largest exposure is in EUR, which accounts for 34,7% of total net assets, followed by USD at 29,6%. Together, these two currencies represent the majority of the Group's

foreign currency exposure and are therefore the main drivers of translation risk.

In addition, Sandvik holds smaller but still relevant exposures in AUD, GBP, and CNY. The category “Other” represents a meaningful proportion of total net assets, indicating that part of the exposure is distributed across several less material currencies.

Overall, Sandvik’s currency exposure is clearly uneven and dominated by a few large positions. This is important for the analysis of hedging strategies, since the largest currencies are likely to have the greatest impact on the Group’s translation risk.

Table 1: Sandvik Net Assets by Foreign Currency in 2025

Currency	SEK bn	Share (%)
EUR	35.14	34.7
USD	29.94	29.6
AUD	9.08	9.0
GBP	5.12	5.1
CNY	3.46	3.4
Other	18.44	18.2
Total	101.18	100.0

Source: Sandvik Annual Report 2025 and own calculations.

4.3 Preliminary Evidence of Fat Tails in Sandvik’s FX Returns

Table 2 reports summary statistics for monthly log returns of EURSEK, USDSEK, AUDSEK, GBPSEK, and CNYSEK over the sample period. Mean returns are close to zero across all currency pairs, with EURSEK being the only pair showing a slightly negative mean (-0.011%), while the remaining pairs display small positive means. The highest mean return is observed for USDSEK (0.043%), followed by AUDSEK (0.038%), CNYSEK (0.037%), and GBPSEK (0.004%).

Return volatility differs noticeably across exchange rates, with USDSEK exhibiting the highest monthly standard deviation (2.324%), followed by CNYSEK (2.094%), while GBPSEK (1.987%) and AUDSEK (1.985%) are nearly identical. In contrast, EURSEK (1.367%) displays the lowest volatility, which is consistent with Sweden’s relatively close economic and financial

linkages to Europe.

Table 2: Descriptive Statistics of Monthly FX Returns Against SEK

Currency pair	Mean (%)	Monthly vol. (%)	Skewness	Excess kurtosis
EURSEK	−0.011	1.367	−0.547	1.071
USDSEK	0.043	2.324	−0.238	0.308
AUDSEK	0.038	1.985	−0.151	0.638
GBPSEK	0.004	1.987	−0.245	0.285
CNYSEK	0.037	2.094	−0.317	0.794

Source: ECB, FRED, Investing.com, and own calculations. Monthly log returns over the period April 2009 to March 2026.

The higher-order moments reveal important differences in the shape of the return distributions. Skewness is negative across all currency pairs, indicating that large depreciations of foreign currencies relative to SEK tend to occur more frequently than equally large appreciations. This asymmetry is most pronounced for EURSEK (−0.547) and CNYSEK (−0.317), whereas AUDSEK (−0.151) is the closest to symmetry. USDSEK (−0.238) and GBPSEK (−0.245) display similar and moderate negative skewness. Excess kurtosis is positive for all pairs, indicating fatter tails and a higher likelihood of extreme outcomes than under a normal distribution. This is most pronounced for EURSEK (1.071), followed by CNYSEK (0.794), AUDSEK (0.638), USDSEK (0.308), and GBPSEK (0.285). Overall, these results suggest that departures from normality are present across all pairs, and are particularly notable for EURSEK and CNYSEK, thereby motivating the subsequent analysis of tail risk and extreme exchange rate movements relevant for Sandvik’s foreign exchange exposure.

5 Theoretical Framework

This section establishes the theoretical framework used to analyse Sandvik’s currency risk and optimal hedging strategy. It introduces the relevant risk measures, modelling approach, and optimisation techniques that form the basis for the empirical analysis.

5.1 Risk Measures: Value-at-Risk and Expected Shortfall

To evaluate Sandvik’s currency risk, this paper considers two downside risk measures: Value-at-Risk (VaR) and Expected Shortfall (ES). Both measures focus on adverse outcomes and are

therefore more suitable than standard variance-based measures when the objective is to assess potential losses arising from exchange rate movements.

For a loss variable L and confidence level α , Value-at-Risk is defined as:

$$\text{VaR}_\alpha = \inf\{v \in \mathbb{R} : F_L(v) \geq \alpha\} \quad (2)$$

where $F_L(v)$ denotes the cumulative distribution function of losses. Intuitively, VaR_α gives the loss threshold that will not be exceeded with probability α . For example, a 95% VaR of SEK 100 million means that losses are expected to remain below this level in 95% of cases, while larger losses may occur in the remaining 5%.

Expected Shortfall is defined as:

$$\text{ES}_\alpha = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha] \quad (3)$$

and measures the average loss conditional on losses exceeding the VaR threshold. This representation holds under continuous loss distributions. This is satisfied in the Monte Carlo framework applied in Section 5.2 which approximates an underlying continuous-time model. While VaR identifies the cutoff point of the loss distribution, ES captures the severity of losses beyond that point. Under a standard normal loss distribution at the 95% confidence level, ES exceeds VaR by a factor of approximately 1.25, under fat-tailed distributions this ratio is higher, providing a direct empirical test for the presence of non-normality in the simulated portfolio losses. In the presence of fat tails, VaR may understate the risk of extreme outcomes because it ignores the shape of the tail beyond the selected quantile. ES is therefore often considered more informative in settings where large losses are rare but potentially severe (Yamai & Yoshida, 2005).

In the optimisation literature (Rockafellar & Uryasev, 2000), Expected Shortfall is often referred to as Conditional Value-at-Risk (CVaR). Under the continuous loss distributions generated by our Monte Carlo simulation, ES and CVaR are equivalent. The CVaR-minimisation problem introduced in Section 5.4.2 can therefore be interpreted as directly minimising Expected Shortfall.

Coherence provides another important criterion for evaluating risk measures. Artzner et al. (1999) define a coherent risk measure as one that satisfies four properties: monotonicity, sub-

additivity, positive homogeneity, and translation invariance. The most relevant property in this context is subadditivity, which means that the measured risk of a combined portfolio should not exceed the sum of the risks of its individual parts.

VaR does not always satisfy this property, especially when losses are not normally distributed. This matters for currency portfolios, where returns may be skewed, fat-tailed, or exposed to joint extreme movements. ES, by contrast, satisfies the coherence properties and is therefore better suited for measuring portfolio tail risk. For Sandvik, where total currency risk depends on both individual exposures and the way currencies move together, this distinction is relevant. If several currencies move adversely at the same time, a VaR-based analysis may understate the severity of the portfolio's downside risk.

5.2 Monte Carlo Simulation with Student- t Copula

To model Sandvik's currency risk, this paper uses a Monte Carlo simulation framework with a Student- t copula. The purpose is to simulate a large number of possible one-year exchange-rate outcomes for Sandvik's main currency exposures. Compared with a Gaussian specification, the Student- t framework is more flexible because it allows for fat-tailed return distributions and joint extreme currency movements. This is important when estimating downside risk using VaR and Expected Shortfall.

The simulation is based on monthly exchange-rate returns estimated over a 17-year historical window and then converted into annualised input parameters. For each currency i , the marginal return distribution is modelled as a Student- t distribution with parameters (μ_i, σ_i, ν_i) . Here, μ_i is the estimated mean return, σ_i is the volatility parameter that determines the dispersion of simulated returns around the mean, and ν_i is the degrees-of-freedom parameter governing tail thickness. Economically, σ_i captures how much the currency tends to fluctuate in normal periods, while ν_i captures how likely extreme movements are relative to a normal distribution. Lower values of ν_i imply heavier tails and a higher probability of large exchange-rate movements.

Dependence across currencies is modelled separately through a Student- t copula. Let R denote the correlation matrix estimated from historical returns, and let ν_c denote the copula degrees

of freedom. Correlated shocks are drawn from a multivariate Student- t distribution:

$$Z_j \sim t_{\nu_c}(0, R), \quad j = 1, \dots, N, \quad (4)$$

where N is the number of simulations. In this paper, $N = 100,000$.

Each simulated shock is transformed into a uniform variable using the univariate Student- t distribution function:

$$U_{ij} = F_{t, \nu_c}(Z_{ij}), \quad (5)$$

and then mapped into a simulated return through the inverse marginal distribution:

$$r_{ij} = F_i^{-1}(U_{ij}), \quad (6)$$

where F_i^{-1} is the inverse CDF of the fitted marginal distribution for currency i . Because F_i is fitted using (μ_i, σ_i, ν_i) , this step incorporates the estimated mean, volatility, and tail thickness of each currency while preserving the dependence structure imposed by the copula.

The simulated end-of-period exchange rate for currency i in simulation j can be written as:

$$S_{iT}^{(j)} = S_{i0} \exp(r_{ij}), \quad (7)$$

where S_{i0} is the initial exchange rate. Since Sandvik's exposures e_i are measured in SEK at the initial exchange rate, the simulated SEK value of each exposure is obtained by scaling the initial exposure by the simulated exchange-rate return. The simulated unhedged portfolio value is therefore:

$$V^{(j)} = \sum_{i=1}^n e_i \exp(r_{ij}). \quad (8)$$

The empirical distribution of $V^{(j)}$ is then used to compute portfolio losses. In each simulation, the loss is defined as the difference between the expected portfolio value and the simulated portfolio value:

$$L^{(j)} = \mathbb{E}[V] - V^{(j)}. \quad (9)$$

VaR and ES are then estimated from the simulated loss distribution.

For comparison, the analysis also includes a Gaussian copula benchmark based on the same

exposure structure and correlation matrix. Under a Gaussian copula, extreme losses in different currencies are less likely to occur simultaneously because tail dependence is zero. By contrast, the Student- t copula allows extreme currency movements to cluster, which can materially increase simulated downside risk. Comparing the Gaussian and Student- t specifications therefore isolates the effect of fat tails and tail dependence on the estimated risk measures and hedging strategy.

5.3 Currency Correlation

For multinational companies like Sandvik, the overall exposure to currency risk is not only a function of individual currency volatilities, but also of the relationships between currencies. Positive correlation implies that currencies tend to move in the same direction, increasing portfolio risk when exposures are aligned. Conversely, low or negative correlations reduce risk by providing natural diversification. Ignoring correlation effects would therefore overstate diversification benefits or underestimate portfolio risk. The correlation matrix for the chosen currencies is reported in Table 3.

Table 3: Correlation Matrix for Key Currency Pairs (vs SEK)

	EUR	USD	AUD	GBP	CNY
EUR	1.000	0.572	0.226	0.582	0.529
USD	0.572	1.000	0.425	0.595	0.908
AUD	0.226	0.425	1.000	0.326	0.459
GBP	0.582	0.595	0.326	1.000	0.618
CNY	0.529	0.908	0.459	0.618	1.000

Source: Own calculations based on monthly log returns from Riksbank exchange-rate data, 2009-04 to 2026-03

The strongest linear dependence is observed between USDSEK and CNYSEK (0.908), reflecting the managed nature of the CNY/USD exchange rate. In contrast, AUDSEK displays notably low correlations with the other pairs, particularly EURSEK (0.226), suggesting it offers natural diversification benefits within the portfolio. Overall, currency correlation serves as a foundational element in both the Monte Carlo simulations and the optimization of hedging strategies presented in the following sections.

5.4 Optimisation Framework: VaR-Optimal vs. CVaR-Optimal Hedging

This section presents the two optimisation frameworks compared in the analysis. Both are formulated so that every term in the objective function is denominated in SEK billions. This ensures that the cost-sensitivity parameter λ is dimensionless in both models and carries an identical economic interpretation: at $\lambda = 1$, one SEK billion of additional annual hedge cost is considered equivalent to one SEK billion of additional risk. The appropriate value of λ for any given firm is not an analytical output but a governance input, reflecting the board's or senior management's internal cost-to-risk preferences.

5.4.1 VaR-Optimal Hedging

The VaR-optimal framework minimises the 95th-percentile portfolio loss, penalised by the annual cost of the hedge:

$$\min_{\mathbf{h} \in [0,1]^5} \underbrace{\text{VaR}_{95\%}(\mathbf{h})}_{\text{SEK bn}} + \lambda \cdot \underbrace{\mathbf{h}^\top \mathbf{c}}_{\text{SEK bn}} \quad (10)$$

where $\text{VaR}_{95\%}(\mathbf{h}) = \inf\{v \in \mathbb{R} : P(L(\mathbf{h}) \leq v) \geq 0.95\}$ is estimated as the 95th percentile of simulated portfolio losses, $L_j(\mathbf{h}) = \mathbb{E}[V(\mathbf{h})] - V_j(\mathbf{h})$ is the loss in Monte Carlo scenario j , and $\mathbf{c} = \mathbf{e} \circ \boldsymbol{\delta}$ is the annual cost vector with e_i denoting the net asset exposure and $\delta_i = (S_i - F_i)/S_i$ the per-unit forward premium for currency i .

5.4.2 CVaR-Optimal Hedging

The CVaR-optimal framework, equivalently ES, directly targets the expected loss in the worst 5% of scenarios:

$$\min_{\mathbf{h} \in [0,1]^5} \underbrace{\text{CVaR}_{95\%}(\mathbf{h})}_{\text{SEK bn}} + \lambda \cdot \underbrace{\mathbf{h}^\top \mathbf{c}}_{\text{SEK bn}} \quad (11)$$

where $\text{CVaR}_{95\%}(\mathbf{h}) = \mathbb{E}[L(\mathbf{h}) \mid L(\mathbf{h}) \geq \text{VaR}_{95\%}(\mathbf{h})]$ and λ carries the same interpretation as in Equation 10. CVaR is estimated using the Rockafellar and Uryasev (2000) formula:

$$\text{CVaR}_{95\%}(\mathbf{h}) \approx \min_{\zeta} \left\{ \zeta + \frac{1}{(1-\alpha)N} \sum_{j=1}^N \max(L_j(\mathbf{h}) - \zeta, 0) \right\} \quad (12)$$

where $\alpha = 0.95$ and ζ represents the VaR threshold. The term $\max(L_j(\mathbf{h}) - \zeta, 0)$ captures losses above this threshold, so the expression accounts for the severity of losses in the worst 5% of simulated outcomes.

6 Data

This section describes the empirical inputs used to estimate Sandvik's currency risk. The analysis combines exchange-rate data, foreign-currency net asset exposures, and forward-rate estimates to construct the Monte Carlo simulation and hedging scenarios.

6.1 Market Data

Exchange rate data are sourced from the Swedish Riksbank's monthly average series. The sample covers 202 monthly observations from April 2009 to March 2026, yielding the 17-year estimation window used for all statistical parameters. The five currencies modelled EUR, USD, AUD, GBP, and CNY correspond to Sandvik's five largest individually disclosed foreign-currency net asset positions, which together account for approximately 81.8% of total foreign-denominated net assets, with the remaining in the category "Other", as reported in the 2025 Annual Report.

Sandvik's net asset exposures are derived from the 2025 Annual Report and yields: SEK 35.14 bn in EUR, SEK 29.94 bn in USD, SEK 9.08 bn in AUD, SEK 5.12 bn in GBP, SEK 3.46 bn in CNY, and SEK 18.44 bn in Other for a total modelled exposure of SEK 101.180 bn (see Table 1).

One-year forward rates are computed from covered interest rate parity (CIP):

$$F_{1Y,i} = S_{0,i} \cdot \frac{1 + r_{\text{SEK}}}{1 + r_i}, \quad (13)$$

using the Riksbank policy rate of 1.75% as r_{SEK} and each currency's prevailing central bank rate as of March 2026 as r_i . The resulting spot rates, forward rates, and implied hedge costs are presented in Table 4.

Table 4: Spot Rates, 1-Year Forward Rates, and Hedge Costs (March 2026)

Currency	S_0	r_i (%)	r_{SEK} (%)	F_{1Y}	δ_i (%)
EURSEK	10.691	2.00	1.75	10.665	0.25
USDSEK	9.223	3.64	1.75	9.054	1.82
AUDSEK	6.515	4.10	1.75	6.368	2.26
GBPSEK	12.326	3.75	1.75	12.088	1.93
CNYSEK	1.338	3.10	1.75	1.321	1.31

Source: Riksbank monthly averages, central bank policy rates as of March 2026, own calculations

The hedge cost pattern is economically intuitive. All foreign currencies carry policy rates above the Riksbank rate, so forward rates are uniformly below spot and hedging is costly for all five pairs. AUD carries the highest implied hedge cost at 2.26% of notional, reflecting the RBA² cash rate of 4.10%, while EUR is the cheapest to hedge at just 0.25% given the ECB deposit rate of 2.00% is closest to the Riksbank rate of 1.75%. GBP and USD carry intermediate costs of 1.93% and 1.82% respectively, driven by the Bank of England rate of 3.75% and the FED rate of 3.64%. CNY sits at 1.31%, reflecting the PBOC³ rate of 3.10%. This cost structure has direct implications for the optimization in Section 7, as EUR is the least costly currency to hedge despite representing Sandvik’s largest single foreign exposure at SEK 35.14 bn.

The hedge cost used in the model should be interpreted as a simplified mid-market proxy rather than the actual transaction price available to Sandvik. In practice, banks quote forward contracts with bid-ask spreads and margins reflecting liquidity, counterparty credit risk, administrative costs, and dealer profit. Since Sandvik-specific dealer quotes are not publicly observable, these bank margins are not included in the baseline cost estimates. The optimisation therefore uses CIP-implied hedge costs as lower-bound estimates of actual implementation costs.

The annualised simulation inputs are reported in Table 5. These summarise the historical mean returns, volatilities, estimated marginal degrees of freedom, and Sandvik’s currency exposures used in the Monte Carlo analysis.

Table 5: Annualised Monte Carlo Input Parameters

Currency	Ann. mean μ_i (%)	Ann. vol. σ_i (%)	MLE ν_i	Exposure (SEK bn)
EURSEK	−0.14	4.73	10.8	35.14
USDSEK	0.51	8.05	18.8	29.94
AUDSEK	0.45	6.88	8.9	9.08
GBPSEK	0.05	6.88	17.2	5.12
CNYSEK	0.45	7.25	8.7	3.46
Total modelled exposure				82.74

Source: Riksbank, Sandvik Annual Report 2025, and own calculations.

²Reserve Bank of Australia

³People’s Bank of China

7 Analysis

This section applies the simulation framework to Sandvik’s currency exposures. It first estimates the unhedged VaR and ES, then evaluates selected hedging cases and compares the VaR- and CVaR-optimal hedge ratios.

7.1 Assumptions for the Analysis

The analysis is based on the data and simulation inputs described in Section 6. To keep the model manageable, the following assumptions are imposed.

First, Sandvik’s foreign-currency net asset positions are assumed to remain constant over the one-year horizon. The model therefore captures translation risk from exchange-rate movements, but does not account for changes in the underlying net asset positions due to acquisitions, divestments, retained earnings, or intra-year balance sheet adjustments.

Second, the analysis focuses on Sandvik’s five largest individually disclosed currency exposures: EUR, USD, AUD, GBP, and CNY. These account for approximately 82% of total foreign-denominated net assets. The residual “Other” category is excluded from the simulation because it is not available at the individual currency level.

Third, exposures are measured in SEK at the initial exchange rate. Simulated portfolio values are therefore obtained by scaling the initial SEK exposure by the simulated exchange-rate return, rather than by modelling the foreign-currency notional directly.

Fourth, hedging is implemented using one-year forward contracts. The annual hedge cost for each currency is proxied by the forward discount relative to the spot rate:

$$\delta_i = \frac{S_{0,i} - F_{1Y,i}}{S_{0,i}},$$

where $S_{0,i}$ is the initial spot rate and $F_{1Y,i}$ is the one-year forward rate. This proxy captures the interest-rate differential implied by covered interest rate parity, but excludes bid-ask spreads, credit charges, liquidity premia, and other transaction costs.

Finally, historical means, volatilities, correlations, and Student- t parameters estimated over

the April 2009–March 2026 window are assumed to be informative for the forward-looking simulation period. The Student- t copula specification is treated as the primary model, while the Gaussian copula is retained as a benchmark for comparison.

7.2 Baseline: Unhedged Portfolio

Table 6: Baseline Risk Metrics: Unhedged Portfolio

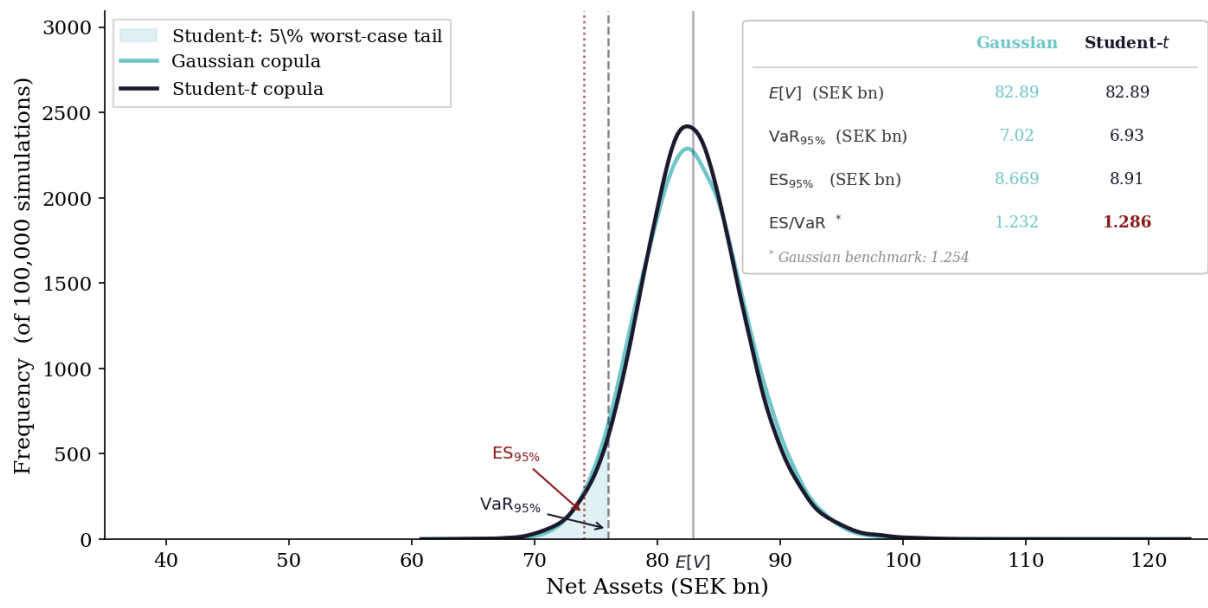
Metric	Gaussian MC	Student- t MC
Expected net assets (SEK bn)	82.89	82.89
VaR _{95%} (SEK bn)	7.02	6.93
ES _{95%} (SEK bn)	8.669	8.91
ES / VaR ratio	1.232	1.286
<i>Analytical Gaussian benchmark: ES / VaR $\approx$ 1.254.</i>		
<i>Ratio > 1.254 under Student-t is consistent with fat-tail effects beyond Gaussian.</i>		
<i>Source: Own Monte Carlo simulations</i>		

Sandvik’s top-five foreign currency positions total SEK 82.74 bn, representing 81.8% of total foreign-denominated net assets. Under both distributional specifications, the Monte Carlo simulation produces an expected net asset value of approximately SEK 82.89 bn, reflecting near-zero mean log-returns across all five currency pairs over the full 2009–2026 estimation window. At the 95% confidence level, the Gaussian copula yields a VaR of SEK 7.02 bn and an ES of SEK 8.669 bn, while the Student- t copula produces a slightly lower VaR of SEK 6.93 bn but a higher ES of SEK 8.91 bn.

The difference is important: the Student- t VaR is 1.28% below its Gaussian counterpart, whereas the Student- t ES is 2.78% above it. The pattern is consistent with fat-tailed distributions concentrating probability mass in the extreme left tail without necessarily shifting the 95th percentile. The ES / VaR ratios provide a formal diagnostic for fat-tail presence. Under the Gaussian copula, the ratio is 1.232, marginally below the analytical Gaussian benchmark of 1.254, attributable to portfolio diversification across five currencies and finite-sample simulation noise. Under the Student- t copula, the ratio rises to 1.286, exceeding the Gaussian benchmark and confirming that Sandvik’s currency portfolio exhibits tail risk beyond what a normal distribution would predict.

As Figure 2 illustrates, the distributional difference between the two specifications is concentrated entirely in the tails: the bulk of the distribution is near-identical, while the left tail under the Student- t copula extends materially further. The expected shortfall of SEK 8.91 bn represents the average loss in the worst 5% of scenarios, and it is this figure, not the VaR, that the CVaR-optimal hedging strategy is designed to minimise.

Figure 2: Baseline Probability Distribution: Unhedged Portfolio - Gaussian vs Student- t Copula.



Source: Own Monte Carlo simulations based on Riksbank monthly exchange-rate data, Sandvik Annual Report 2025 exposure data, and CIP-implied forward rates. 100,000 simulations.

7.3 Distributional Comparison: Gaussian vs. Student- t

The divergence between the two distributional assumptions is most pronounced in the tails, which is precisely where it matters most for risk management. Under the Gaussian copula, VaR and ES differ by a factor of 1.232, broadly consistent with the analytical benchmark of 1.254. Under the Student- t copula, this ratio rises to 1.286, indicating that the average loss in the worst 5% of scenarios is proportionally larger than a Gaussian model would suggest.

Critically, the two specifications yield similar VaR estimates (SEK 7.02 bn vs. SEK 6.93 bn, a difference of only 1.28%), but their ES estimates diverge more substantially (SEK 8.669 bn vs. SEK 8.91 bn, a difference of 2.78%). This asymmetry illustrates the well-known limitation of VaR: it is relatively insensitive to the tail specification because it depends only on the loss at

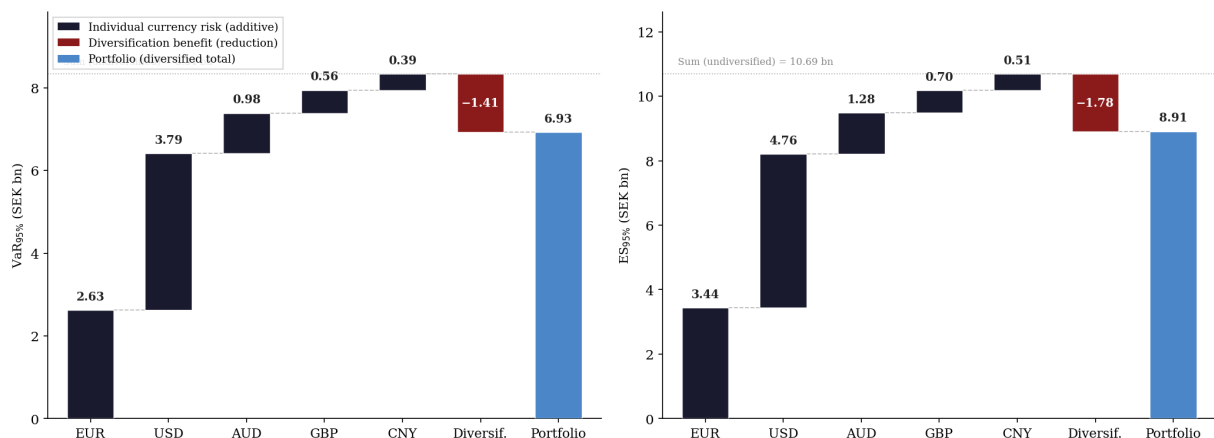
a single quantile, whereas ES integrates over the entire tail and therefore amplifies any distributional difference that lives beyond the threshold.

For the five currencies in Sandvik’s portfolio, two marginal distributions exhibit clear fat-tail behaviour. AUDSEK and CNYSEK converge to low degrees of freedom under maximum likelihood estimation ($\hat{\nu}_{\text{AUD}} = 8.9$ and $\hat{\nu}_{\text{CNY}} = 8.7$), indicating materially heavier tails than a normal distribution. EURSEK also produces moderate fat tails ($\hat{\nu} = 10.8$), while GBPSEK sits in a similar range ($\hat{\nu} = 17.2$). USDSEK is the most Gaussian of the five ($\hat{\nu} = 18.8$), reflecting a more stable long-run exchange rate relationship. This heterogeneity motivates examining how the two optimisation frameworks respond differently when the largest tail-risk positions are targeted for hedging.

7.4 Individual Currency Risk Decomposition

Figure 3 decomposes Sandvik’s portfolio VaR and ES into individual currency contributions, using the Student-*t* Monte Carlo as the primary model.

Figure 3: Individual Currency Risk Decomposition – Student-*t* MC.



Source: Own Monte Carlo simulations based on Riksbank monthly exchange-rate data, Sandvik Annual Report 2025 exposure data, and CIP-implied forward rates.

Table 7 below summarises the individual currency contributions. USD represents the largest single risk driver despite EUR being the largest exposure in nominal terms: the individual ES attributable to USDSEK alone is SEK 4.76 bn, compared to SEK 3.44 bn for EURSEK. This occurs because USD carries substantially higher annualised exchange-rate volatility (8.05% versus EUR’s 4.73%), so that despite a smaller nominal exposure, its per-unit fluctuations translate

into disproportionately large tail losses. The combined sum of undiversified individual risks is SEK 10.69 bn for ES, the diversification effect reduces this to SEK 8.91 bn, a benefit of SEK 1.78 bn.

Importantly, the currency ranking changes slightly between VaR and ES. USD remains the dominant risk contributor under both measures, but the relative ES contribution of AUDSEK is proportionally larger than its VaR contribution, producing an individual ES/VaR ratio of 1.306. This reflects AUD's fat-tailed marginal distribution ($\hat{\nu} = 8.9$) and is precisely the kind of information that VaR alone would conceal.

Table 7: Individual Currency Risk Contributions (Student- t MC)

Currency	Exposure (SEK bn)	Indiv. VaR (SEK bn)	Indiv. ES (SEK bn)	ES/VaR
EURSEK	35.14	2.63	3.44	1.307
USDSEK	29.94	3.79	4.76	1.255
AUDSEK	9.08	0.98	1.28	1.306
GBPSEK	5.12	0.56	0.70	1.250
CNYSEK	3.46	0.39	0.51	1.307
Sum (undiversified)	82.74	8.35	10.69	—
Diversification benefit	—	1.41	1.78	—
Portfolio (diversified)	82.74	6.93	8.91	1.286

Source: Own Monte Carlo simulations based on Riksbank monthly exchange-rate data, Sandvik Annual Report 2025 exposure data, and CIP-implied forward rates.

7.5 Hedging Case 1: 100% Hedge of Largest Exposure (EUR)

The first hedging case addresses the most natural question a corporate treasury faces: what happens if we fully hedge our largest foreign currency position? Sandvik's EUR exposure of SEK 35.14 bn is the single largest among the five currencies, representing 42% of total modelled exposure. Table 8 reports the effect of fully hedging this position using a one-year forward contract at the CIP-implied forward rate of EURSEK 10.665.

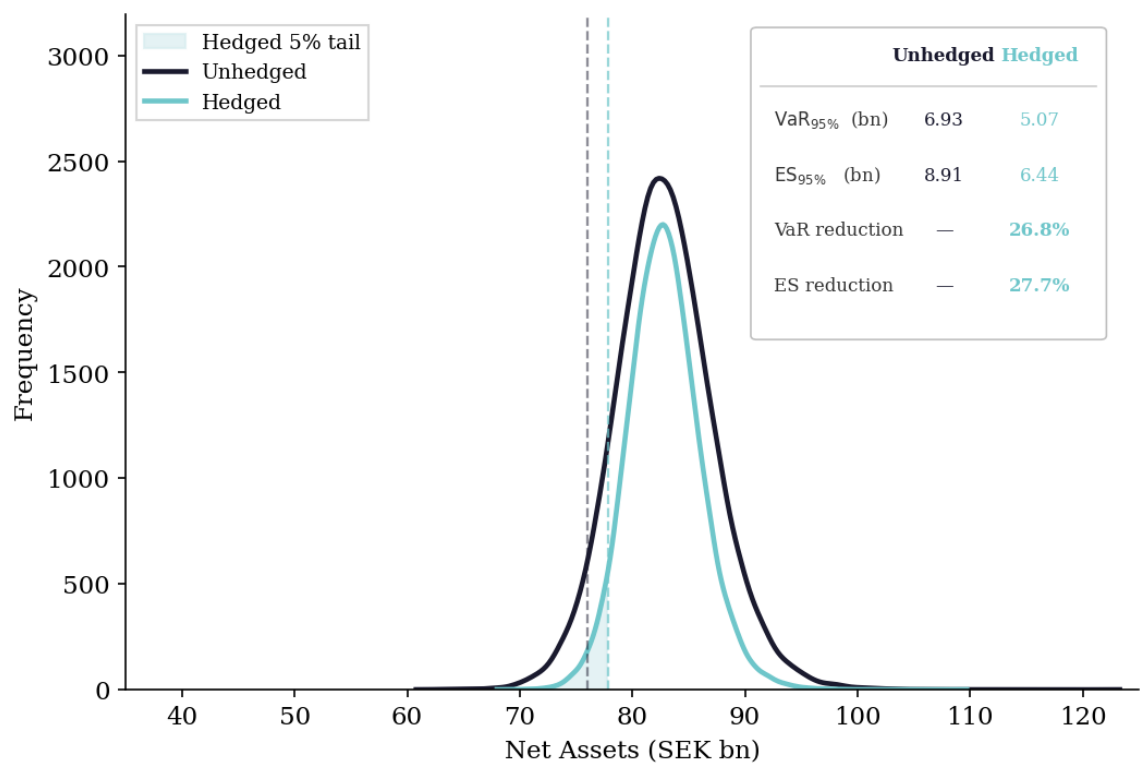
Table 8: Hedging Case 1: 100% EUR Hedge (Student- t MC)

Metric	Baseline (Unhedged)	Case 1 (100% EUR)
Expected net assets (SEK bn)	82.89	82.80
VaR _{95%} (SEK bn)	6.93	5.07
ES _{95%} (SEK bn)	8.91	6.44
VaR reduction (%)	—	26.8
ES reduction (%)	—	27.7
ES / VaR ratio	1.286	1.271

Source: Own calculations. EUR hedge cost: 0.25% of notional (SEK 0.088 bn p.a.).

Fully hedging EUR reduces VaR by 26.8% and ES by 27.7%, delivering a substantial and broadly symmetric reduction in both risk measures. The ES / VaR ratio decreases modestly from 1.286 to 1.271, reflecting that EUR itself carries moderate fat-tail behaviour ($\hat{\nu} = 10.8$); removing it slightly improves the tail-shape profile of the residual portfolio. The EUR hedge is exceptionally inexpensive: at a forward premium of only 0.25% of notional, the total hedge cost is SEK 0.088 bn per year, delivering an ES reduction of SEK 2.47 bn, a cost efficiency of approximately 28.1 SEK of expected shortfall eliminated per SEK of annual hedge premium.

Figure 4: Hedging Case 1: Portfolio Loss Distribution – 100% EUR Hedge (Student-*t* MC).



Source: Own Monte Carlo simulations based on Riksbank monthly exchange-rate data, Sandvik Annual Report 2025 exposure data, and CIP-implied forward rates.

The intuition is straightforward: EUR is large enough in nominal terms that even its moderate per-unit risk (individual ES/VaR = 1.307) translates into a dominant aggregate contribution. A size-based selection criterion identifies EUR as the primary hedging target because portfolio risk is, in the first instance, a function of exposure magnitude.

7.6 Hedging Case 2: 100% Hedge of Largest Tail-Risk Contributor (USD)

Case 2 selects the hedging target based on a fundamentally different criterion: not the largest nominal exposure, but the largest contributor to portfolio tail risk as measured by individual ES. From Table 7, USDSEK has the highest individual ES among all five currencies (SEK 4.76 bn), exceeding even EUR’s contribution (SEK 3.44 bn) despite carrying a smaller nominal exposure. USD’s outsize risk footprint arises from its substantially higher exchange-rate volatility (8.05% annualised versus EUR’s 4.73%), which means each unit of USD exposure contributes disproportionately more to the portfolio’s tail distribution.

Table 9: Hedging Case 2: 100% USD Hedge (Student- t MC)

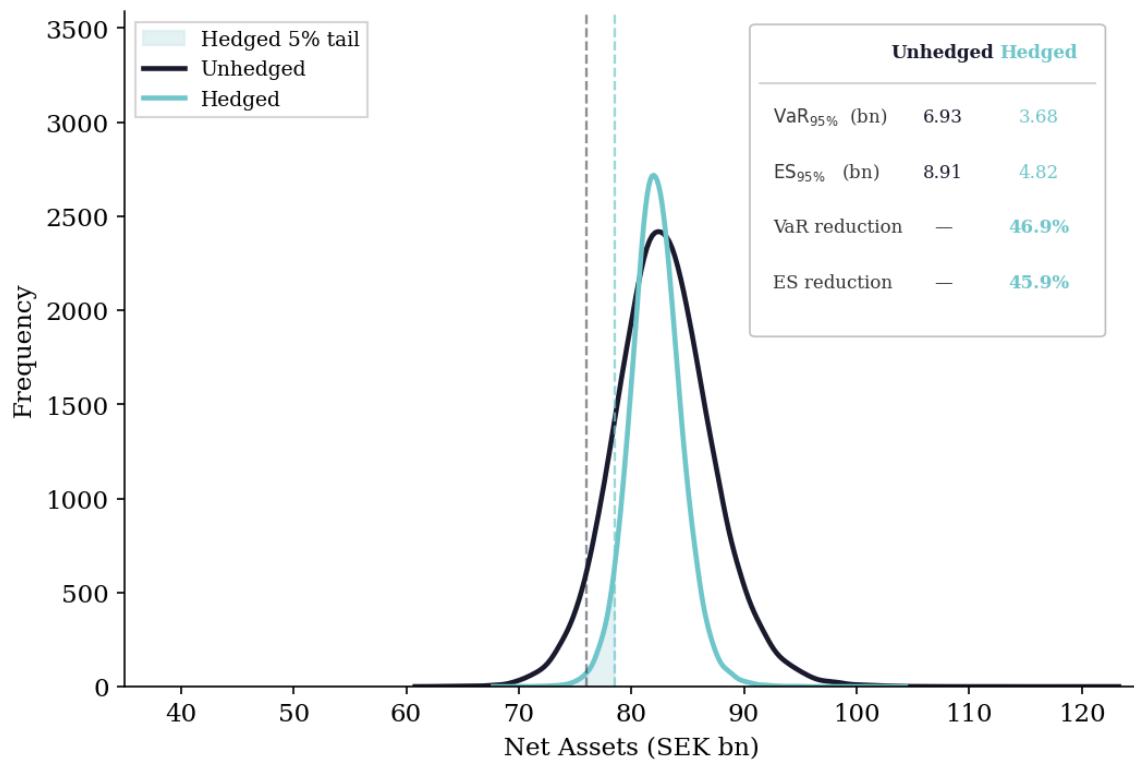
Metric	Baseline (Unhedged)	Case 2 (100% USD)
Expected net assets (SEK bn)	82.89	82.35
VaR _{95%} (SEK bn)	6.93	3.68
ES _{95%} (SEK bn)	8.91	4.82
VaR reduction (%)	—	46.9
ES reduction (%)	—	45.9
ES / VaR ratio	1.286	1.311

Source: Own calculations. USD hedge cost: 1.82% of notional (SEK 0.545 bn p.a.).

Fully hedging USD reduces VaR by 46.9% and ES by 45.9%, nearly double the reduction achieved in Case 1. However, USD carries a materially higher forward premium: at 1.82% of notional, the annual hedge cost is SEK 0.545 bn, approximately 6.2 times the cost of the EUR hedge. On a cost-efficiency basis, the USD hedge removes SEK 4.09 bn of expected shortfall for SEK 0.545 bn of annual premium, a ratio of approximately 7.5:1, substantially below EUR's ratio of 28.1:1. This comparison illustrates an important duality: absolute risk reduction and cost efficiency are distinct objectives. Hedging USD maximises total tail-risk elimination; hedging EUR is more efficient per unit of cost.

The ES / VaR ratio increases slightly from 1.286 to 1.311 after hedging USD. This reflects an important compositional effect: USD is the most Gaussian of the five currencies ($\hat{\nu} = 18.8$), meaning it contributes primarily through size rather than tail shape. Removing USD leaves a residual portfolio dominated by AUD ($\hat{\nu} = 8.9$), EUR ($\hat{\nu} = 10.8$) and CNY ($\hat{\nu} = 8.7$), which are collectively fatter-tailed on a per-unit basis. The ES/VaR ratio rises accordingly, even though the absolute ES falls dramatically.

Figure 5: Hedging Case 2: Portfolio Loss Distribution – 100% USD Hedge (Student- t MC).



Source: Own Monte Carlo simulations based on Riksbank monthly exchange-rate data, Sandvik Annual Report 2025 exposure data, and CIP-implied forward rates.

Comparing Cases 1 and 2 exposes a fundamental tension in FX risk management: *the largest position is not necessarily the largest risk*. EUR is the natural first hedge from a treasury management perspective – it dominates the balance sheet and its forward market is the most liquid. But from a pure tail-risk perspective, USD is the single currency that, if hedged alone, would halve the portfolio’s tail exposure. This distinction is invisible to a VaR-based framework that focuses on quantile thresholds, but it emerges directly from the ES decomposition. The practical implication for Sandvik is that a portfolio manager targeting absolute ES reduction would prioritise USD before EUR, accepting a higher forward premium in exchange for dramatically greater risk elimination.

7.7 VaR-Optimal Hedge Ratios

Table 10 presents the VaR-minimising hedge ratios across the λ grid. At each grid point the optimiser selects the hedge ratio vector that minimises $\text{VaR}_{95\%}(\mathbf{h}) + \lambda \mathbf{h}^\top \mathbf{c}$, with λ representing the marginal rate at which one SEK billion of VaR may be exchanged for one SEK billion of annual cost saving.

Table 10: VaR-Optimal Hedge Ratios Across λ (Student- t MC)

	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 7$	$\lambda = 8$	$\lambda = 10$	$\lambda = 12$
EUR hedge ratio	100%	100%	100%	100%	100%	100%	99%
USD hedge ratio	100%	100%	97%	0%	0%	0%	0%
AUD hedge ratio	100%	100%	0%	0%	0%	0%	0%
GBP hedge ratio	100%	100%	0%	0%	0%	0%	0%
CNY hedge ratio	100%	100%	100%	100%	72%	72%	0%
Portfolio VaR _{95%} (SEK bn)	0.000	0.000	1.337	4.677	4.769	4.769	5.035
Portfolio ES _{95%} (SEK bn)	0.000	0.000	1.741	5.929	6.058	6.058	6.403
Annual hedge cost (SEK bn)	0.981	0.981	0.664	0.131	0.119	0.119	0.086

Source: Own simulations via VaR-minimisation (Eq. 10).

The VaR-optimal strategy exhibits a distinct threshold structure across five phases. In the first phase ($\lambda \leq 2$), the strategy remains at the full-hedge vertex with portfolio VaR and ES equal to zero. As long as full hedging eliminates the 95th-percentile loss entirely, the objective provides no incentive to reduce any hedge ratio: doing so would create residual VaR from zero, which the cost saving cannot offset.

In the second phase ($\lambda = 5-6$), AUD and GBP drop to zero simultaneously while EUR and CNY remain at 100 % and USD falls only marginally to 97 %. AUD and GBP carry the highest forward premiums in the portfolio (2.26 % and 1.93 % respectively) and their contribution to the 95th-percentile threshold is modest relative to that cost, making them the first to exit. EUR, despite being Sandvik's largest exposure at SEK 35.14 bn, remains fully hedged because its forward premium of 0.25 % is by far the lowest in the portfolio.

In the third phase ($\lambda = 7$), USD is dropped entirely while EUR and CNY both remain at 100 %. The residual VaR rises sharply to SEK 4.677 bn and ES to SEK 5.929 bn at an annual cost of just SEK 0.131 bn. This transition reflects the VaR objective's indifference to tail severity: once USD's marginal cost exceeds its contribution to the 95th-percentile threshold, it exits abruptly regardless of the tail risk it leaves behind.

In the fourth phase ($\lambda = 8-10$), CNY is partially reduced to 72 %, while EUR continues to hold at 100 %. The residual VaR plateaus at SEK 4.769 bn and ES at SEK 6.058 bn, un-

changed between $\lambda = 8$ and $\lambda = 10$, reflecting the discrete structure of VaR optimisation: once the portfolio settles at EUR plus partial CNY, no further currency reduction affects the quantile threshold within this range.

In the fifth and final phase ($\lambda \geq 12$), CNY is dropped entirely, leaving only near-full EUR (99 %) as the sole remaining hedge. Both VaR and ES plateau at SEK 5.035 bn and SEK 6.403 bn respectively, at an annual cost of just SEK 0.086 bn. EUR survives at extreme cost sensitivity because its forward premium of 0.25 % remains below the marginal VaR-reduction value that $\lambda = 12$ implies

7.8 CVaR-Optimal Hedge Ratios

Table 11 presents the CVaR-minimising hedge ratios across the same λ grid. The contrast with Table 10 constitutes the central empirical finding of the paper.

Table 11: CVaR-Optimal Hedge Ratios Across λ (Student- t MC)

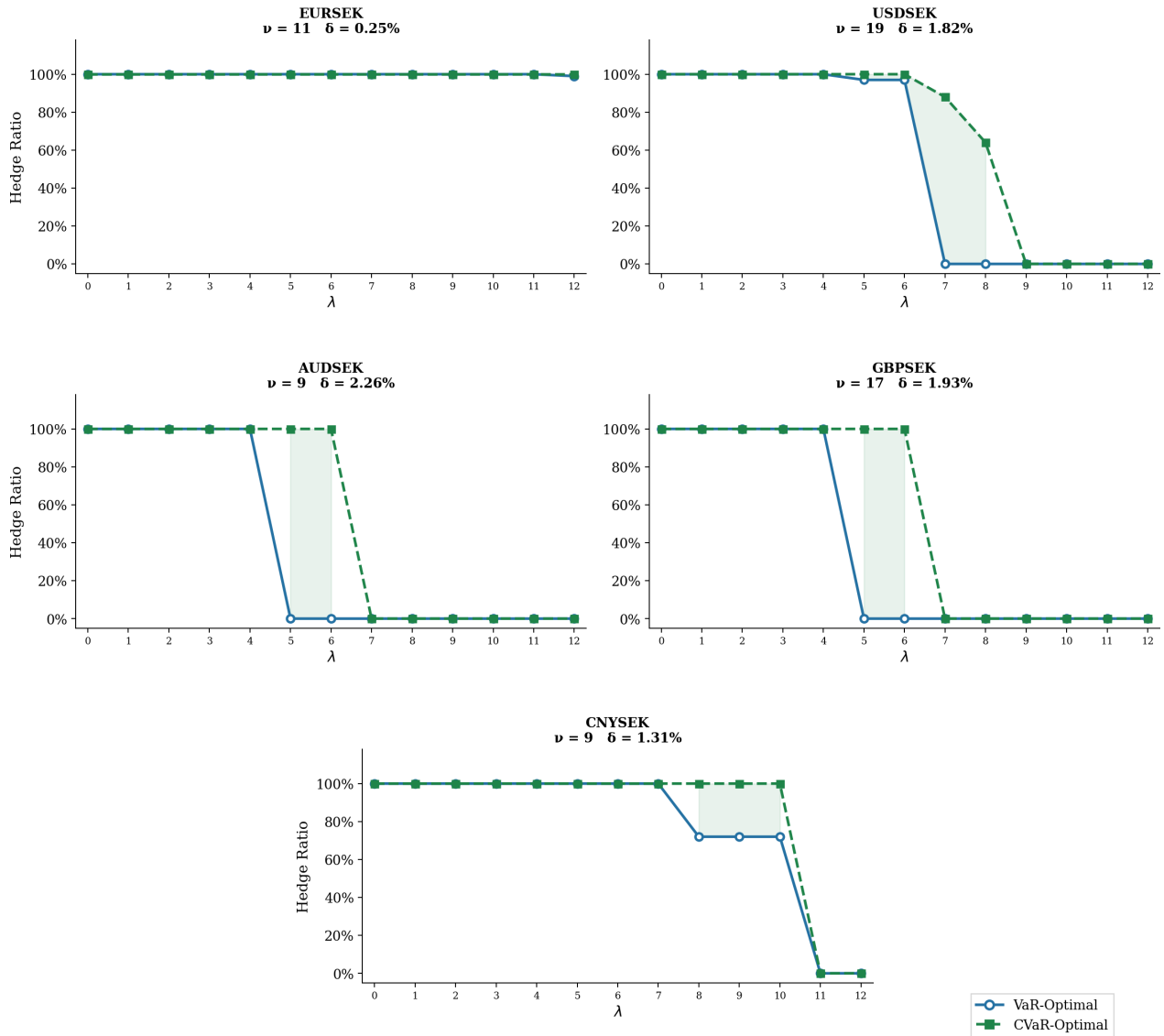
	$\lambda = 0$	$\lambda = 2$	$\lambda = 5$	$\lambda = 7$	$\lambda = 8$	$\lambda = 10$	$\lambda = 12$
EUR hedge ratio	100%	100%	100%	100%	100%	100%	100%
USD hedge ratio	100%	100%	100%	88%	64%	0%	0%
AUD hedge ratio	100%	100%	100%	0%	0%	0%	0%
GBP hedge ratio	100%	100%	100%	0%	0%	0%	0%
CNY hedge ratio	100%	100%	100%	100%	100%	100%	0%
Portfolio VaR _{95%} (SEK bn)	0.000	0.000	0.000	1.606	2.379	4.669	5.027
Portfolio ES _{95%} (SEK bn)	0.000	0.000	0.000	2.076	3.054	5.924	6.393
Annual hedge cost (SEK bn)	0.981	0.981	0.981	0.611	0.478	0.131	0.086

Source: Own simulations via CVaR-minimisation (Eq. 11). Same simulation set as Table 10.

The CVaR-optimal strategy maintains complete hedging of all five currencies for every $\lambda \leq 6$, achieving zero residual VaR and ES throughout. This extends the full-hedge zone by two grid points beyond the VaR-optimal model, which first reduces coverage at $\lambda = 5$. The CVaR objective applies a higher implicit cost to leaving fat-tail currencies unhedged than the VaR objective does, because it integrates over all losses beyond the 95th-percentile boundary rather than simply locating it. Full coverage therefore remains optimal at $\lambda = 5$ and $\lambda = 6$, precisely the range where VaR-optimal has already dropped AUD and GBP.

The transition at $\lambda = 7$ is abrupt and reflects the corner-solution structure of the CVaR objective. AUD and GBP exit simultaneously to 0 %, but USD is reduced only to 88 % rather than eliminated: its larger nominal exposure (SEK 29.94 bn) and higher annualised volatility (8.05 %) ensure that even a partial hedge continues to reduce CVaR materially. At $\lambda = 8$, USD is further reduced to 64 % as cost pressure increases. CNY remains at 100 % through $\lambda = 10$, three grid points beyond the VaR-optimal model's CNY exit. Its low forward premium ($\delta = 1.31$ %) and fat-tailed marginal distribution ($\hat{\nu} = 8.7$) mean its cost-to-ES-contribution ratio remains favourable longer than for any other currency except EUR.

Figure 6: Hedge Ratios: VaR-Optimal vs CVaR-Optimal across λ .



Source: Own Monte Carlo simulations based on Riksbank monthly exchange-rate data, Sandvik Annual Report 2025 exposure data, and CIP-implied forward rates.

At $\lambda = 12$, CNY finally exits, and the CVaR-optimal portfolio converges to EUR-only, the same configuration as the VaR-optimal model. At this point the two strategies produce nearly identical outcomes: residual ES of SEK 6.393 bn (CVaR-optimal) versus SEK 6.403 bn (VaR-optimal) at an annual cost of SEK 0.086 bn each. The divergence between the two frameworks is therefore concentrated in the mid-range $\lambda = 5$ through $\lambda = 10$, where the two objectives produce materially different hedging recommendations and materially different levels of residual tail exposure.

Figure 6 visualises the divergence across all λ values for all five currencies. The USD panel most clearly captures the methodological difference: the VaR-optimal ratio drops to zero at

$\lambda = 7$ while CVaR-optimal holds at 88 %, only beginning to decline at $\lambda = 8$. The CNY panel shows the complementary result: CVaR-optimal retains full CNY coverage through $\lambda = 10$ while VaR-optimal reduces it to 72 % at $\lambda = 8$ and drops it entirely at $\lambda = 12$.

7.9 Head-to-Head Comparison: VaR-Optimal vs. CVaR-Optimal

Tables 10 and 11 together reveal five distinct phases. In the first phase ($\lambda \leq 2$), both frameworks produce identical full-hedge portfolios. In the second phase ($\lambda = 5\text{--}6$), VaR-optimal begins dropping coverage while CVaR-optimal remains fully hedged. In the third phase ($\lambda = 7$), both models have reduced hedging substantially but the CVaR-optimal portfolio retains materially more tail protection, this is the point of peak ES divergence. In the fourth phase ($\lambda = 8\text{--}10$), further unwinding occurs under both models, with the gap narrowing. In the fifth phase ($\lambda \geq 12$), both models converge to EUR-only hedging and produce near-identical outcomes. Table 12 presents the comparison at $\lambda = 7$, the point where the ES gap between the two strategies is at its widest.

Table 12: Strategy Comparison at $\lambda = 7$: Peak ES Gap (Student- t MC)

Metric	Unhedged	VaR-Optimal	CVaR-Optimal
EUR hedge ratio	—	100%	100%
USD hedge ratio	—	0%	88%
AUD hedge ratio	—	0%	0%
GBP hedge ratio	—	0%	0%
CNY hedge ratio	—	100%	100%
VaR _{95%} (SEK bn)	6.931	4.677	1.606
ES _{95%} (SEK bn)	8.911	5.929	2.076
Annual hedge cost (SEK bn)	—	0.131	0.611
ES reduction vs unhedged (SEK bn)	—	2.982	6.835
ES reduction per SEK of cost	—	22.76	11.19

Source: Own Monte Carlo simulations based on Riksbank monthly exchange-rate data, Sandvik Annual Report 2025 exposure data, and CIP-implied forward rates.

At $\lambda = 7$, the two strategies have arrived at fundamentally different portfolio configurations despite sharing the same cost-sensitivity parameter. The VaR-optimal portfolio retains only EUR (100 %) and full CNY (100 %), having dropped USD entirely, at an annual cost of SEK 0.131 bn, and carries a residual ES_{95%} of SEK 5.929 bn. The CVaR-optimal portfolio additionally retains

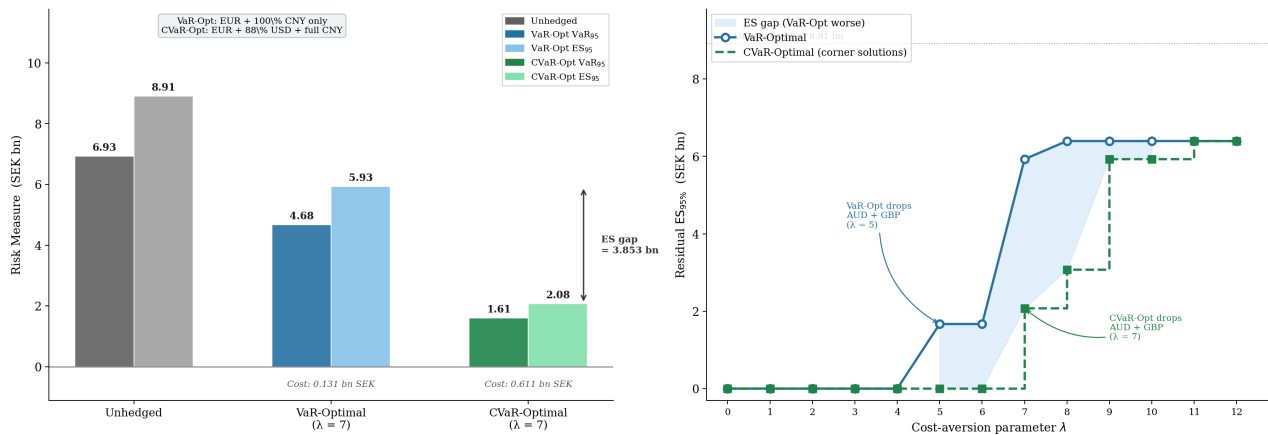
88 % USD and full CNY (100 %), at an annual cost of SEK 0.611 bn, and achieves a residual $ES_{95\%}$ of SEK 2.076 bn. The ES gap between the two strategies is SEK 3.853 bn: at identical λ , CVaR-optimal leaves SEK 3.853 bn less in expected tail loss.

The cost-efficiency row in Table 12 deserves careful interpretation. VaR-optimal achieves a substantially higher ES-reduction-per-SEK ratio (22.76 versus 11.19). This is not because VaR-optimal is more effective at reducing tail risk; it reflects the fact that VaR-optimal is spending very little (SEK 0.131 bn), while CVaR-optimal is spending approximately 4.7 times as much (SEK 0.611 bn) to achieve a much deeper reduction. The high ratio under VaR-optimal is driven entirely by EUR, whose 0.25 % forward premium delivers large absolute risk reduction at minimal cost. Once EUR is secured, the VaR-optimal model stops. The CVaR-optimal model continues to add USD and CNY because their expected tail contributions, weighted over the worst 5 % of scenarios, still justify the incremental cost. This is the core of the paper’s finding: the VaR objective is satisfied once the 95th-percentile threshold has been addressed, whereas the CVaR objective continues to assign value to currencies that contribute disproportionately to the severity of losses beyond that point.

The convergence at $\lambda \geq 12$ is equally informative. Once cost pressure is extreme enough, both models independently arrive at the same conclusion: only EUR remains worth hedging, because its 0.25 % forward premium is the only one that still clears the cost-risk threshold at $\lambda = 12$. The ES difference narrows to essentially zero (SEK 6.393 bn versus SEK 6.403 bn), and the annual hedge cost is identical at SEK 0.086 bn. The divergence between the two frameworks is therefore not a permanent feature of the models, but a phenomenon that is concentrated in the range $\lambda = 5$ through $\lambda = 10$, where cost pressure is sufficient to force trade-offs but not yet so extreme that all hedges are abandoned. It is precisely this middle range that is most relevant for corporate treasury practice.

Figure 7 illustrates the comparison across both dimensions: the left panel shows the full risk profile at $\lambda = 7$, the point of peak ES divergence, and the right panel traces the ES paths for $\lambda \in [0, 12]$ with the gap between the two models shaded throughout.

Figure 7: Head-to-Head: VaR-Optimal vs CVaR-Optimal ($\lambda = 7$ snapshot and full ES frontier, $\lambda \in [0, 12]$).



Source: Own Monte Carlo simulations based on Riksbank monthly exchange-rate data, Sandvik Annual Report 2025 exposure data, and CIP-implied forward rates.

The right panel of Figure 7 captures the full arc of the comparison. The VaR-optimal ES path rises from $\lambda = 5$ onward as hedge coverage is reduced under the quantile objective. The CVaR-optimal path remains flat at zero through $\lambda = 6$ and then rises in a step-function pattern, reflecting the corner-solution property: coverage is maintained until the marginal cost of a currency precisely offsets its contribution to expected tail loss, at which point it exits discontinuously. The shaded gap is widest at $\lambda = 7$ and narrows at $\lambda = 9$ as CVaR-optimal begins to exit positions, closing entirely at $\lambda = 12$ as both models reach the EUR-only configuration.

This pattern, divergence in the mid-range and convergence at the extremes, suggests that the choice of risk measure matters most precisely when the treasury is operating under moderate cost sensitivity, the realistic operating range for most corporate hedging programmes. The robustness check reported in Appendix A.2 indicates that this divergence is robust to the choice of confidence level: the VaR- and CVaR-optimal strategies continue to diverge at the 97.5% and 99% levels, with the disagreement shifting to higher cost-aversion as the level rises rather than disappearing.

8 Discussion

This paper examines whether the choice between Value-at-Risk (VaR) and Expected Shortfall (ES) leads to materially different hedging recommendations for Sandvik's foreign-denominated net assets. The results show that it does, but mainly under specific conditions. Both frame-

works agree on the broad strategic conclusion: Sandvik's translation exposure is large enough that a fully unhedged position is difficult to justify, and the largest exposures should receive substantial hedge coverage. The difference arises in the marginal hedging decisions, especially when hedging costs are introduced and the model must choose which currencies to retain.

The central result is that the CVaR-optimal strategy remains more conservative than the VaR-optimal strategy at moderate levels of cost sensitivity. At low values of λ , both models recommend full hedging. However, as cost sensitivity increases, the VaR-optimal model begins reducing hedge coverage earlier, while the CVaR-optimal model retains protection for currencies that contribute materially to losses in the tail. This is most visible at $\lambda = 7$, where the VaR-optimal strategy has dropped USD entirely and retains only full EUR and full CNY hedging, whereas the CVaR-optimal strategy still retains 88% USD coverage alongside full EUR and full CNY. The result is a materially lower residual ES under the CVaR-optimal strategy, falling from SEK 5.929 billion to SEK 2.076 billion, although annual hedge cost increases from SEK 0.131 billion to SEK 0.611 billion.

8.1 When Does the Choice of Risk Measure Matter?

The choice of risk measure matters most when three conditions are present: non-normal return distributions, positive dependence across currencies, and a treasury objective that places weight on severe downside outcomes rather than only threshold losses. In a purely Gaussian setting, VaR and ES tend to produce similar rankings because both are largely driven by the same second-moment information. In the Student- t specification, however, the ES/VaR ratio is 1.286, exceeding the Gaussian benchmark of approximately 1.254, indicating that Sandvik's currency portfolio contains more tail risk than a normal model would imply.

This is precisely the setting in which VaR becomes less informative. VaR identifies the loss threshold exceeded in the worst 5% of scenarios, but it does not measure how severe losses are beyond that threshold. ES, by contrast, averages across the tail and therefore assigns more weight to currencies that contribute disproportionately in adverse joint scenarios. The practical implication is not that VaR and ES imply entirely different treasury policies. Rather, the choice of risk measure affects the calibration of the hedge, especially for currencies whose tail-risk contribution is larger than their nominal exposure alone would suggest.

8.2 Why Should Sandvik Hedge Translation Exposure?

Whether Sandvik should hedge translation exposure is not obvious. Translation losses do not directly reduce operating cash flow, and hedging them creates real forward costs. The case for hedging therefore cannot be that all currency volatility should be eliminated, but rather that large translation effects may affect Sandvik's financial flexibility and external risk perception.

This is economically relevant for Sandvik. According to the 2025 Annual Report, Sandvik had SEK 101.1 billion of net assets in foreign subsidiaries. A 5% adverse exchange-rate movement would reduce other comprehensive income by approximately SEK 5.1 billion, mainly through translation exposure in equity. Relative to year-end equity of SEK 93.2 billion, this equals around 5.4% of reported equity. Although unrealised, such losses affect consolidated equity and therefore book leverage, equity ratios, and the balance sheet metrics observed by creditors, rating agencies, and shareholders (Sandvik AB, [2026](#)).

The covenant argument should not be overstated, since Sandvik states that its loan conditions do not currently include financial covenants linked to key figures. The stronger rationale is instead that selective hedging can preserve balance sheet resilience and support Sandvik's financial policy. Sandvik reports a financial net debt/EBITDA ratio of 0.9x against a target below 1.5x, maintains a BBB+ long-term credit rating, and emphasises that a strong balance sheet supports investments, acquisitions, dividends, and shareholder returns (Sandvik AB, [2026](#)).

Thus, translation exposure is not only an accounting issue. It can influence perceived financial strength and strategic flexibility. The implication is not full hedging, but selective hedging where downside balance sheet protection is large relative to hedge cost.

8.3 Implications for Corporate Treasury

For corporate treasury practice, the main implication is that VaR should not be the only measure guiding hedging decisions when currency returns exhibit fat tails and dependence in stress states. A VaR-based framework may appear cost-efficient because it focuses on reducing the loss threshold. However, it can leave more residual exposure in the tail once the threshold itself has been addressed. The CVaR-optimal framework is more conservative because it continues to assign value to hedges that reduce the severity of losses beyond the VaR cutoff.

This does not mean that non-financial firms should mechanically replace VaR with ES. Banks use ES partly because of regulatory capital requirements, while Sandvik manages a corporate balance sheet subject to different objectives, including transparency, operational simplicity, accounting treatment, and internal risk appetite. VaR remains intuitive and easy to communicate. The stronger conclusion is therefore that ES should complement VaR when the portfolio exhibits clear tail risk. In Sandvik's case, ES reveals residual downside exposure that a VaR-based optimisation can underweight.

8.4 Limitations

The analysis has several limitations. First, although the Student- t copula improves on the Gaussian benchmark, it still imposes symmetric tail behaviour. Real FX returns may be asymmetric, especially during market stress or policy-driven currency movements. Second, the estimated degrees of freedom, volatilities, and dependence structure are sensitive to the chosen sample period. Exchange-rate regimes can shift over time, so historical tail behaviour may not persist over the next year. Third, the model assumes a static one-year hedge horizon with no rebalancing, whereas an actual treasury function would adjust hedge ratios as exposures and market conditions change.

There are also measurement limitations. Hedge costs are proxied using the spot-forward differential implied by covered interest parity. This abstracts from the fact that banks must earn a margin when selling hedging instruments. Actual forward rates faced by Sandvik would likely include bid-ask spreads, liquidity premia, counterparty credit charges, operational costs, and dealer profit. The exclusion of these costs means that the analysis may understate the true cost of hedging and therefore overstate optimal hedge ratios. Nevertheless, the simplification is necessary because Sandvik-specific dealer quotes are not publicly available. Finally, ES is more informative than VaR in the tail, but it is also more model-dependent and harder to estimate precisely. Its theoretical advantages therefore come with practical implementation costs.

8.5 Further Extensions

Several extensions could strengthen the analysis. A natural next step would be to allow for time-varying volatility and dependence, for example through a GARCH model or rolling copula estimation. Skewed Student- t marginals could also be introduced to capture asymmetric currency movements. In addition, a dynamic hedging framework with periodic rebalancing would

move the model closer to actual treasury practice.

Future work could also broaden the economic interpretation of the hedge decision. One extension would be to compare forward hedges with currency options, since options provide downside protection while preserving upside potential. Another would be to stress test the portfolio under specific macroeconomic scenarios, such as a sharp SEK appreciation or a global risk-off episode.

9 Conclusion

The research question that this paper set out to answer: *“Does the choice between Value-at-Risk (VaR) and Expected Shortfall (ES) as the primary risk measure lead to materially different hedging recommendations for Sandvik’s foreign-denominated net assets, and if so, what are the implications for corporate treasury risk management?”*

The main conclusion is that the choice of risk measure does matter, but primarily in the calibration of the hedge rather than in the overall strategic direction. Both the VaR- and CVaR-based frameworks indicate that Sandvik is materially exposed to translation risk and that its major foreign-currency exposures should receive substantial hedge coverage. However, once fat tails, currency dependence, and hedging costs are taken into account, the two frameworks produce materially different hedge ratios for some currencies.

The unhedged portfolio illustrates the economic relevance of the problem. Under the Student- t specification, Sandvik’s top-five currency exposures generate a 95% VaR of SEK 6.931 billion and a 95% ES of SEK 8.911 billion. The corresponding ES/VaR ratio of 1.286 exceeds the Gaussian benchmark of approximately 1.254, indicating that the portfolio contains more tail risk than a normal model would imply. This matters because VaR identifies only a loss threshold, whereas ES captures the average severity of losses beyond that threshold.

The central empirical finding is that VaR- and CVaR-based optimisation do not produce identical hedge portfolios. At low levels of cost sensitivity, both approaches recommend full hedging. At moderate cost sensitivity, however, the CVaR-optimal strategy remains more conservative. At $\lambda = 7$, the VaR-optimal strategy has dropped USD entirely and retains only full EUR and full CNY hedging, while the CVaR-optimal strategy additionally retains 88% USD cov-

erage. As a result, residual ES falls from SEK 5.929 billion under the VaR-optimal strategy to SEK 2.076 billion under the CVaR-optimal strategy, although annual hedge cost increases from SEK 0.131 billion to SEK 0.611 billion. The CVaR-optimal strategy therefore leaves SEK 3.853 billion less expected tail loss at the same cost-sensitivity parameter.

At the same time, the difference between the two approaches should not be overstated. The models do not imply opposite treasury policies. EUR remains the most robust hedge because it combines a large exposure with a low forward premium, and both frameworks converge toward similar recommendations when cost sensitivity becomes sufficiently high. The disagreement is concentrated in the middle range, where hedging costs matter but are not so large that all non-core hedges are abandoned. This is also the range most relevant for corporate treasury decisions.

The broader implication is that VaR remains useful, but should not be the only measure guiding hedging decisions when FX returns are fat-tailed and currencies may move jointly in stress scenarios. VaR is simple and easy to communicate, but it provides no information about losses beyond the selected quantile. ES is more model-dependent and harder to estimate, but it gives a more complete picture of downside risk. For Sandvik, the results suggest that an ES-informed framework can reveal residual tail exposure that a VaR-based optimisation may underweight.

In conclusion, Sandvik's foreign-denominated net assets expose the firm to significant translation risk, and the choice between VaR and ES can alter the optimal hedge ratios in a practically meaningful way. The effect is strongest for currencies that contribute disproportionately to tail losses rather than simply being the largest nominal exposures. The answer to the research question is therefore yes: the choice of risk measure can lead to materially different hedging recommendations. For corporate treasury, this means that risk measurement is not merely a technical modelling choice, but part of the strategic design of the hedging policy.

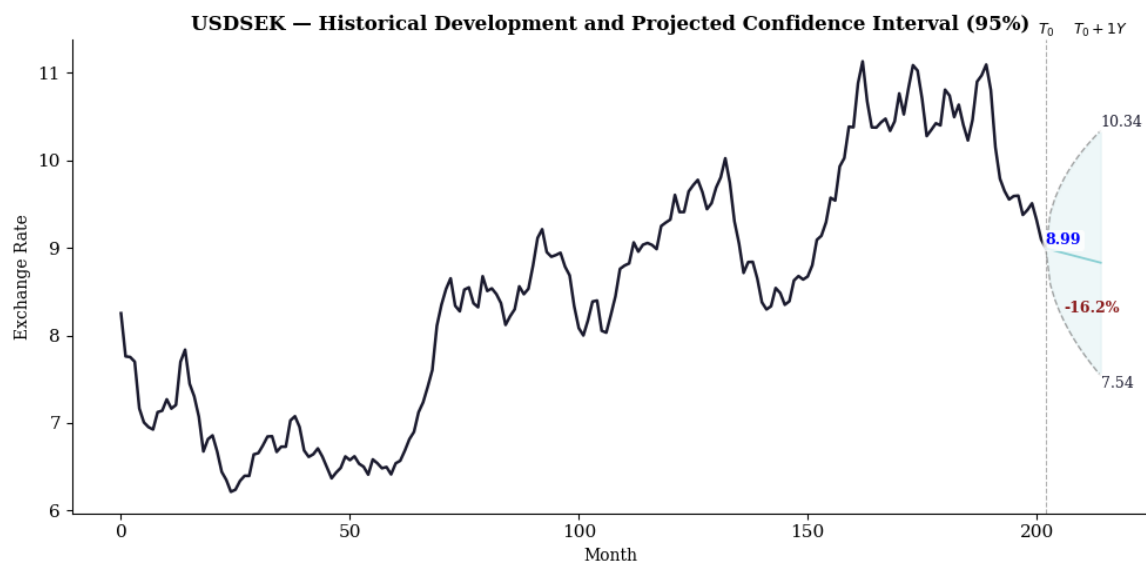
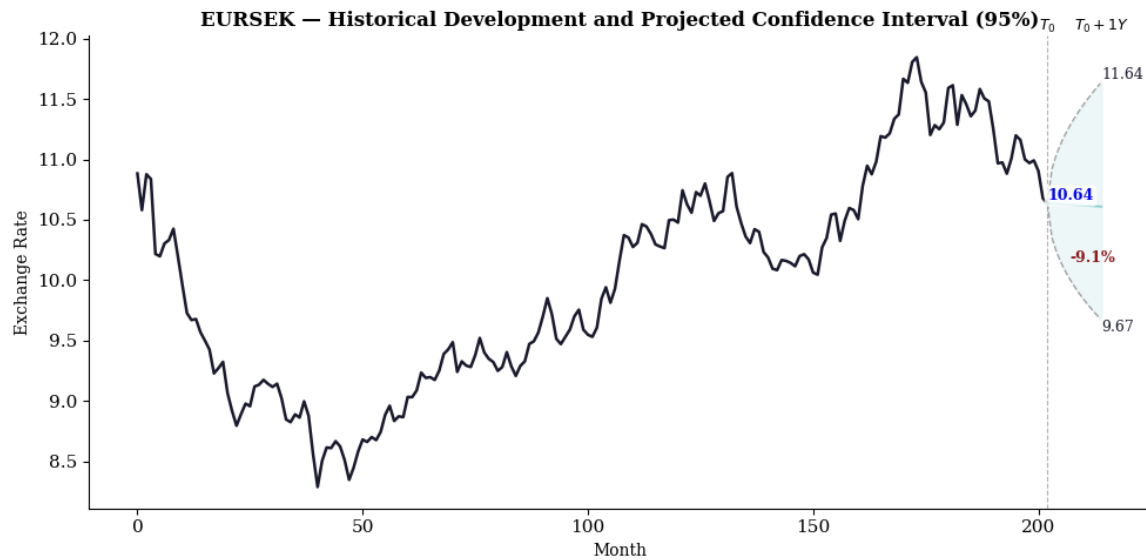
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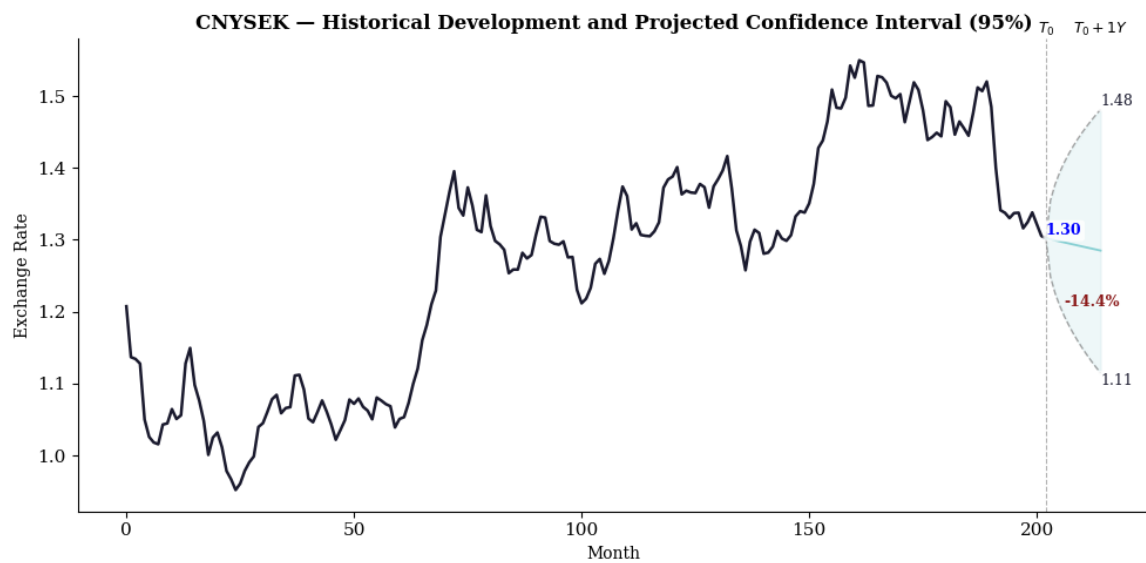
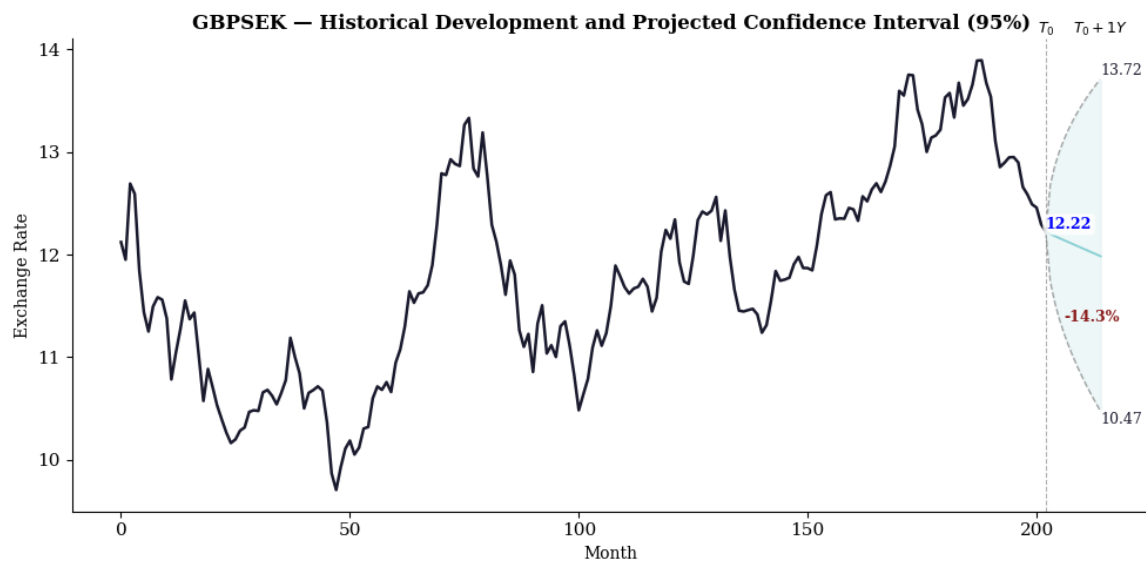
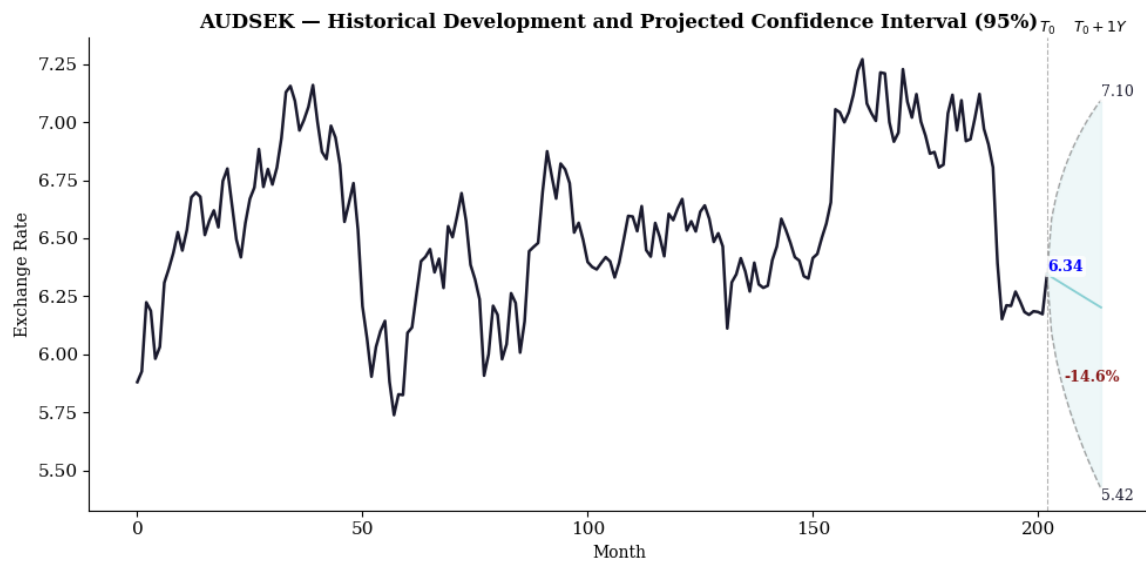
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A Appendix

A.1 Historical FX Development and Confidence Intervals





A.2 Robustness of the Hedging Conclusion to the Confidence Level

The central comparison (Figures 6 and 7) is conducted at the 95% confidence level. To verify that the conclusion about the VaR- versus CVaR-optimal hedge ratios is not specific to this choice, we re-run the full λ -sweep at the 97.5% and 99% levels, holding the Monte Carlo simulation and all inputs fixed. Figure 8 reports the resulting hedge-ratio paths and Table 13 the associated risk metrics.

The qualitative conclusion is robust across all three confidence levels. At every level, EUR remains fully hedged throughout (lowest forward premium); AUD and GBP, which combine high hedge costs with comparatively thin tails, are the first to be dropped under both objectives; and the CVaR-optimal strategy retains the fat-tailed, high-exposure currencies USD and CNY at a higher cost-aversion than the VaR-optimal strategy. The ordering of the two strategies is therefore preserved: CVaR-optimal is consistently more conservative in the tail.

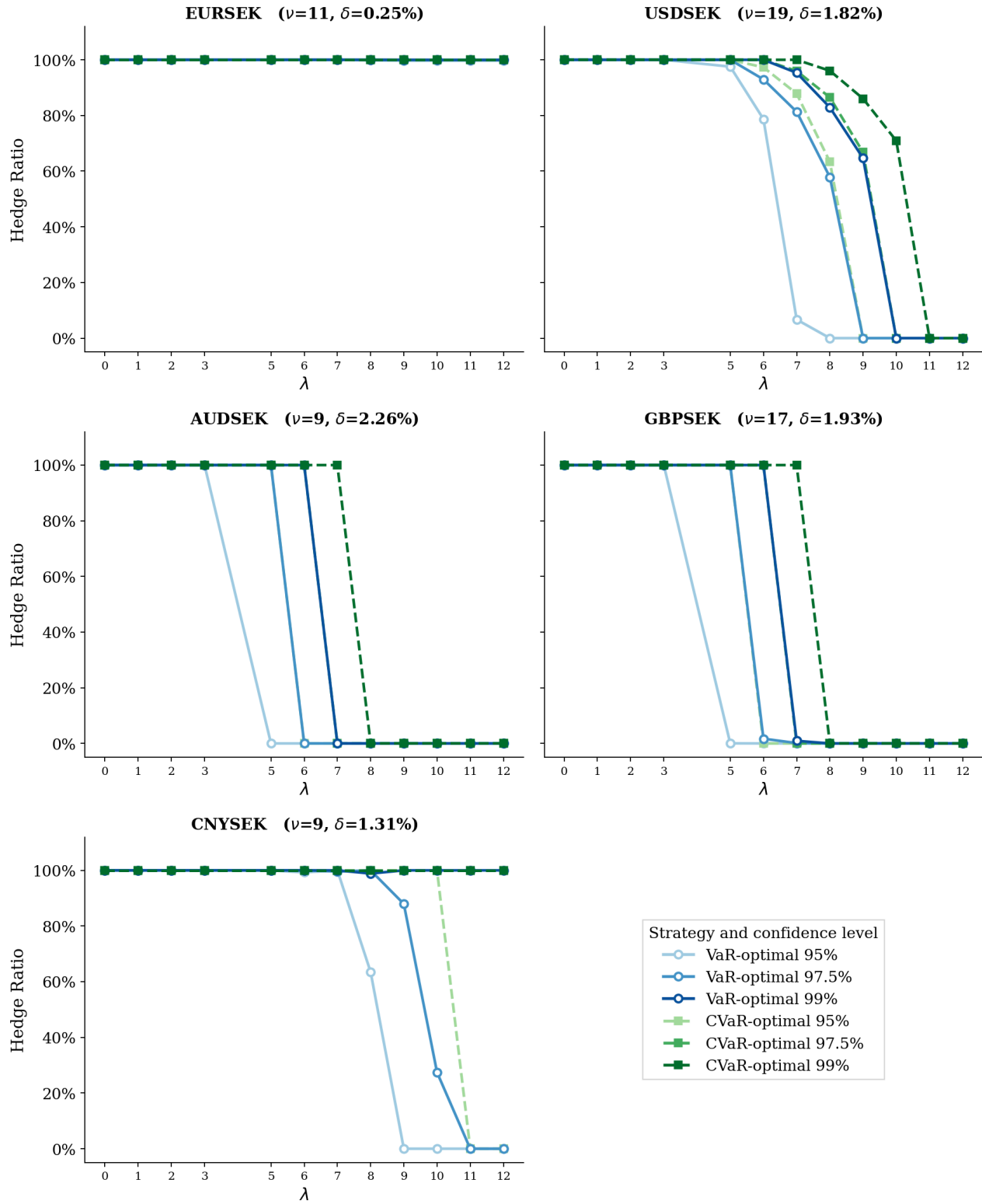
What changes with the confidence level is the *location* of the unwinding, not its direction. As the level rises, more probability mass sits in the tail, so each currency remains worth hedging up to a higher λ and the entire path shifts rightward. The VaR-optimal USD position, the dominant tail contributor, is unwound near $\lambda = 7$ at 95%, near $\lambda = 9$ at 97.5%, and near $\lambda = 10$ at 99% (Figure 8, Table 13) the range over which the two strategies disagree therefore moves to higher cost-aversion rather than narrowing. The residual-ES gap between the two strategies stays economically large, exceeding SEK 3 bn at every level and reaching about SEK 4 bn at 99%. The conclusion that the choice of risk measure materially changes the hedge is thus not an artefact of the 95% threshold, if anything it is sharpest in the most adverse states.

Table 13: Robustness Across Confidence Levels (Student- t MC)

Confidence level	Unhedged ES/VaR	Gaussian benchmark	USD VaR-optimal exit (λ)
95.0%	1.29	1.25	≈ 7
97.5%	1.22	1.19	≈ 9
99.0%	1.18	1.15	≈ 10

Source: Own Monte Carlo simulations based on Riksbank monthly exchange-rate data, Sandvik Annual Report 2025 exposure data, and CIP-implied forward rates.

Figure 8: Robustness of Hedge Ratios to the Confidence Level: VaR-Optimal vs CVaR-Optimal across λ .



Source: Own Monte Carlo simulations based on Riksbank monthly exchange-rate data, Sandvik Annual Report 2025 exposure data, and CIP-implied forward rates.